

VEDIC SHARMA

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EDUCATION

PhD Computer Science

September 2020 - Present

University of Toronto, supervised by Prof. Peter Marbach

Building theoretical foundations in Multi-Agent Reinforcement Learning (MARL). We approach multi-agent systems from a multidisciplinary perspective, incorporating insights from sociology, economics, and reinforcement learning to design systems that emulate human decision-making and collaboration. Our goal is to establish a robust theoretical foundation for multi-agent systems to address fundamental questions such as:

- How can agents optimize collective social welfare through a decentralized system?
- What information should agents share, and how should they use it?
- How can we design multi-agent systems that dynamically adapt to complex environments?

Research Intern

August 2018 - December 2018

Indian Statistical Institute - Supervised by Prof. B Sury

- Studied classical problems in lattice tilings including works of Minkowski, Hajos and Redei. I worked on an alternate proof for Minkowski's theorem (2 dimensions), showing every lattice tiling in 2D space, using unit squares, share a common edge. The method was extended to 3 dimensions as well.
- Explored the latest findings in the field and determined tile invariants for various polyomino tile sets using a coloring technique.

M.Sc. (Hons.) Mathematics and B.E. (Hons.) Computer Science

August 2014 - June 2019

Birla Institute of Technology and Science, Pilani, Hyderabad

CGPA: 9.23 out of 10

Delhi Public School, Bangalore East (Grade 11-12)

July 2012 - June 2014

Overall Percentage: 91.8

Mounds View High School, Minnesota, USA (Grade 9-10)

July 2010 - June 2012

TEACHING ASSISTANT POSITIONS (TA)

TA positions held as a PhD student at the University of Toronto:

Design and Analysis of Data Structures CSCB63H3S (65.0 hours)

Jan 2026 - Apr 2026

Introduction to Computer Science I CSCA08H3S (65.0 hours)

Jan 2026 - Apr 2026

Introduction to the Theory of Computation CSC236H1F (60.0 hours)

Sep 2025 - Dec 2025

Foundations of Computer Science I CSC110H1F (60.0 hours)

Sep 2025 - Dec 2025

Data Structures and Analysis CSC263H1F (60.0 hours)

Sep 2024 - Dec 2024

Social and Information Networks CSC303H1S (60.0 hours)

Jan 2024 - Apr 2024

Data Structures and Analysis CSC263H1F (60.0 hours)

Jan 2024 - Apr 2024

Data Structures and Analysis CSC263H1F (60.0 hours)

Sep 2023 - Dec 2023

Data Structures and Analysis CSC263H1S (60.0 hours)

Jan 2023 - Apr 2023

Introduction to the Theory of Computation CSC236H1F (60.0 hours)

Sep 2022 - Dec 2022

Introduction to Computer Programming CSC108H1S (60.0 hours)

Jan 2022 - Apr 2022

ACADEMIC ACHIEVEMENTS

1. **Department of Computer Science Graduate Program Award** for Fall 2025, and Winter 2026 terms by University of Toronto.

2. **Departmental Fellowship Award** for the year 2022-2023 by University of Toronto.
3. **Rank 1** in the class of 2019 in the M.Sc. Mathematics program in BITS Pilani.
4. **Merit Scholarship** for 5 terms by BITS Pilani, awarded to top 3% of the highest academic standing.
5. **Ranked 1st and 2nd** in ATMOS Mathematics Paper Presentation, in 2015 and 2016 respectively.
6. Achieved an **all India rank 6452 (99.6th percentile)** in JEE Advanced 2014 (IIT Entrance Exam).

COURSES

System Modelling and Analysis Algorithms for Collective Decision
 Number Theory Cryptography Graphs and Networks Algebra Discrete Mathematics
 Algorithm Design Optimization Operations Research Topology Numerical Analysis
 Data Structures and Algorithms Logic in Computer Science Theory of Computation

TECHNICAL PROFICIENCY

C, C++, Python, VB.NET, VBA, Objective-C, Swift, Git
 Algorithm Design, Software Development, Mathematical Modelling

WORK EXPERIENCE

Apple, Hyderabad - Software Engineer *June 2019 - August 2020*

- Gathered user behaviour data from multiple projects, connected to an external server, mapped to user friendly variables, and generated analytics/visualizations (using a third party software) for business teams and executives.

Apple, Hyderabad - Intern *January 2019 - June 2019*

- Designed algorithms to efficiently collect and filter user behaviour data from internal applications.

Goldman Sachs, Bangalore - Engineering Summer Analyst *May 2018 - July 2018*

- Designed a database model for an internal application. The new schema provided all the capabilities which the old one did, as well as remedies of problems being currently faced (eg. long query timings due to many table joins) and problems that could arise in the future.

Bharat Heavy Electrical Limited, Trichy *May 2016 - July 2016*

India's largest power generation equipment manufacturer, owned by the Government of India

- Created a mathematical model to optimize logistics cargo transportation costs. Constraints included the amount of boxes of each type, vehicle weight and size limits, distance travelled by vehicle type. A simpler version of this problem, the bin packing problem is NP-Hard. However, by applying some heuristics, I created an algorithm which gave a sudo-solution in polynomial time.

ACADEMIC PROJECTS

Domination in Circulant Graphs *August 2018 - December 2019*

The study aimed to determine a minimum dominating set of circulant graph, $C_n^{1,m}$ for large values of n . I found that there exists n_0 for every m , such that for $n > n_0$, $\gamma(C_{n+5}^{1,2}) = \gamma(C_n^{1,2}) + 1$ (γ represents the minimum dominating number). The symmetry of the graph as well as the repetition of the states in the corresponding digraph [*Min-Plus Algebra and Graph Domination*] was used to determine patterns in large graphs.

Sequence having the n^{th} differences in a Geometric Progression *February 2016 - July 2016*

Finite differences can determine a continuous function approximating a sequence having the n^{th} differences in an arithmetic progression (I call them ghost AP). The study built upon this method, to

generalize for sequences whose n^{th} differences are in a geometric progression. The first n terms of the sequence were taken and extended as a ghost arithmetic progression. Taking limit as n tends to infinity, we can obtain the formula for determining the n^{th} term. Sum of the sequence, a vector space representing the sequence were also part of the whole analysis on ghost GP. The paper was sent to Mathematical Association of America for the Mathematics Magazine. The decision was received on 2 November 2017: the “paper is too long and too formal for an expository journal like Mathematics Magazine” and was not published.

Generating Course Timetable for the University

December 2017 - April 2018

Designed an algorithm to create a course timetable for BITS Pilani. The timetable illustrates which classes are conducted when, where and by whom. The modified version of the genetic algorithm used, had a fitness function with ten parameters, and a crossover function which created multiple timetables (for next generation) with a single fit parent.

An Alternate Understanding of Infinity

July 2016 - September 2016

The presentation was part of a competition (**ATMOS Paper Presentation**), held annually in the university, open to all undergraduates in the country. The presentation was about an alternate definition of infinity, namely $\infty = c(\mathbb{N})$. All other “infinities” are a derivative of the newly defined ∞ . The advantages and the drawbacks to this method of defining infinity were also discussed. **Placed 2nd**

OTHER WORK, EXPERIENCES, AND ACHIEVEMENTS

Machine Learning C++, Python

September 2017 - April 2018

- Digit recognition, analysis of various linear models, concept learning, decision trees

Wizard’s Chess

January 2017 - April 2017

Built a real life chess game in which the moves played by the player is automatically recognized by reed switches hidden inside the board. The corresponding output, calculated by a combination of python programs and a chess engine (Stockfish), is shown in a LCD display and voiced out by a speaker

Data Mining C++

September 2016 - November 2016

- Tested diabetic patient (FP-Growth), association rules (Apriori) to determine political party

Computer Graphics C

August 2016 - November 2016

- Particle-based viscoelastic fluid simulation, Bezier curves, 2D to 3D object creation, scene creation, tree drawings

Eagle Scout (Boy Scouts of America)

January 2012